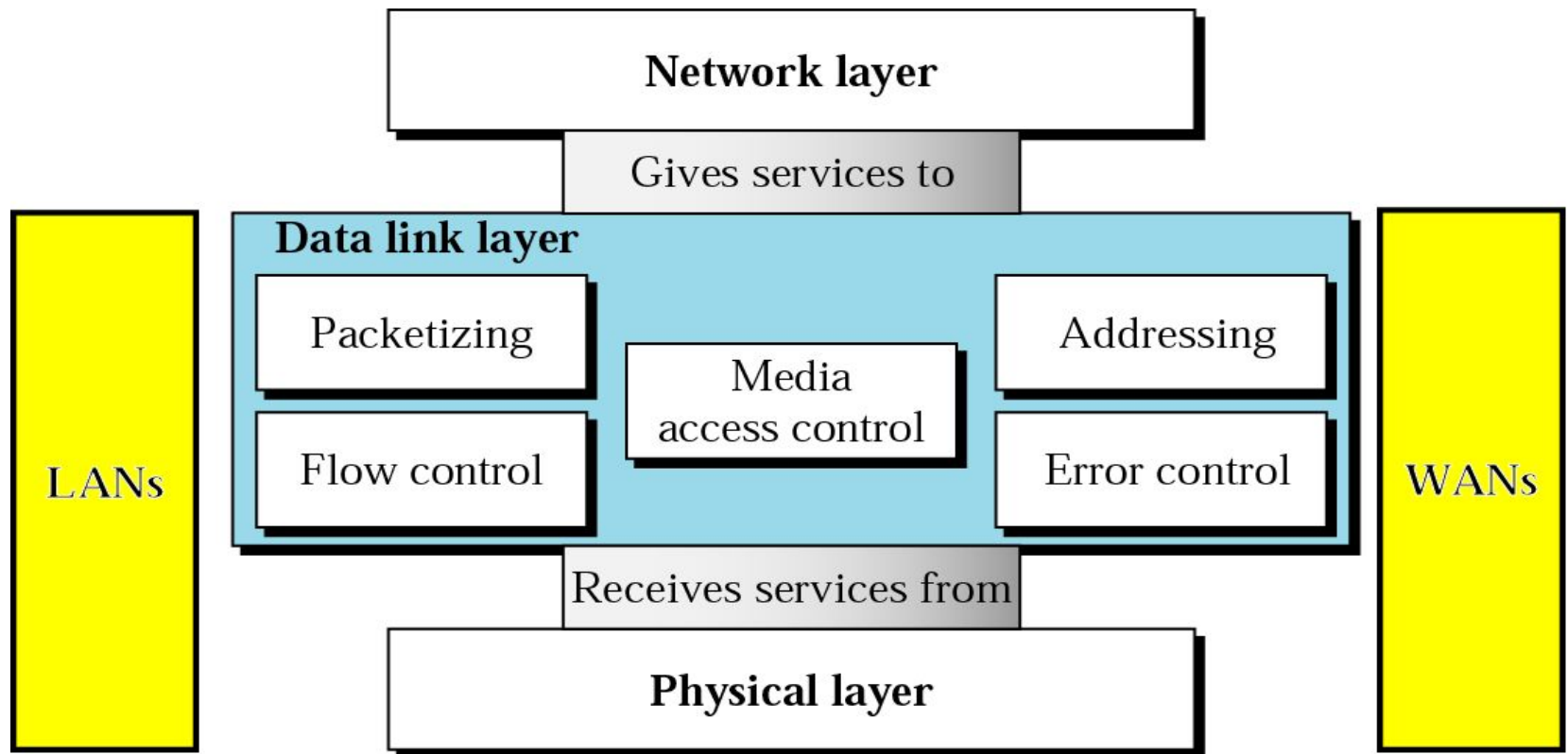


Computer Communication & Networks

Lecture 17

Datalink Layer: Multiple Access

Data Link Layer



Data Link Layer Topics to Cover

Error Detection and Correction

Data Link Control and Protocols

Multiple Access

Local Area Networks

Wireless LANs

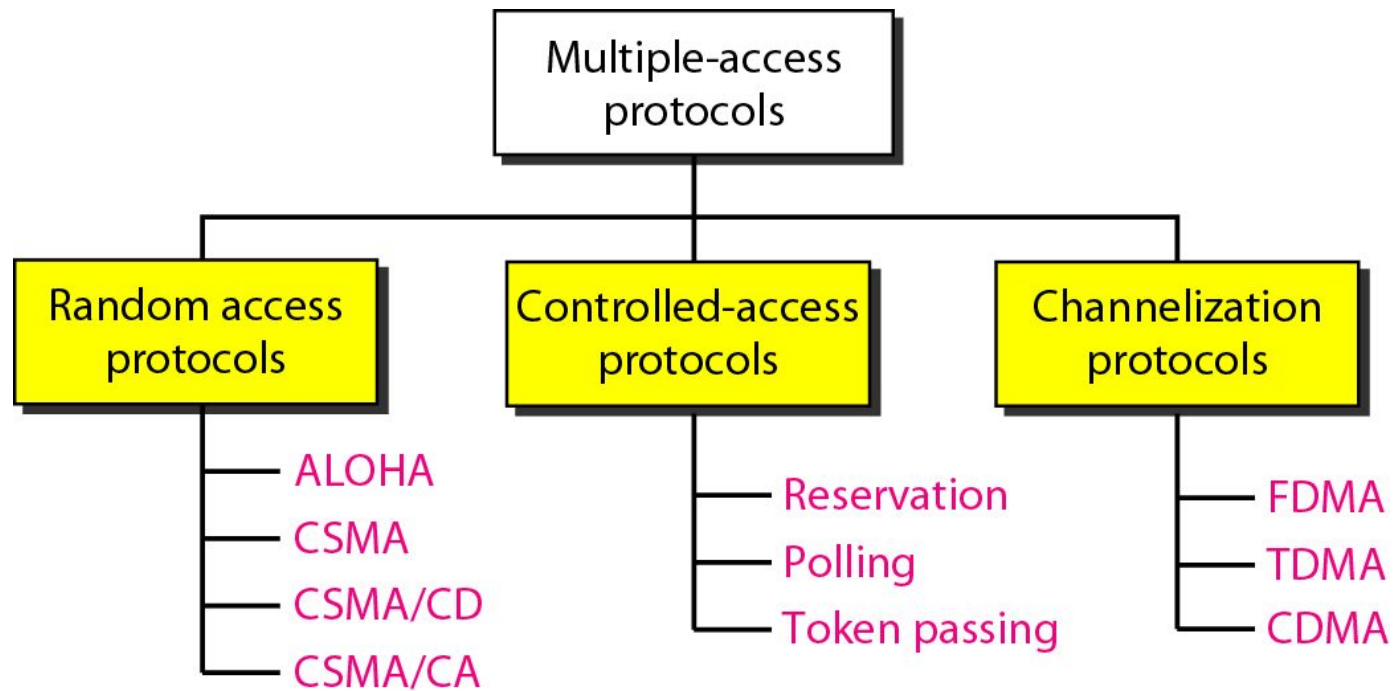
Multiple access problem

- Example:
- Cocktail party – many people gather together in a large room
- Broadcast medium – air
- Another example: a classroom
- Human protocols:
 - “Give everyone a chance to speak”
 - “Don’t speak until you are spoken to”
 - “Don’t monopolize the conversation”
 - “Raise your hand if you have a question”
 - “Don’t interrupt when someone is speaking”
 - “Don’t fall asleep when someone else is talking”

Multiple access protocols

- In LANs, WiFi, satellite networks, cocktail party
- If more than 2 users send @ the same time - collision
- All collided packets are lost -> waste of bandwidth
- Ideally, the MAC protocol for a broadcast channel with the bit-rate R bps should satisfy:
 - if only 1 node is sending than the throughput is R
 - when M nodes have data to send than the throughput is R/M
 - decentralized protocol – no master
 - simple & inexpensive to implement

Taxonomy of Multiple-Access Protocols

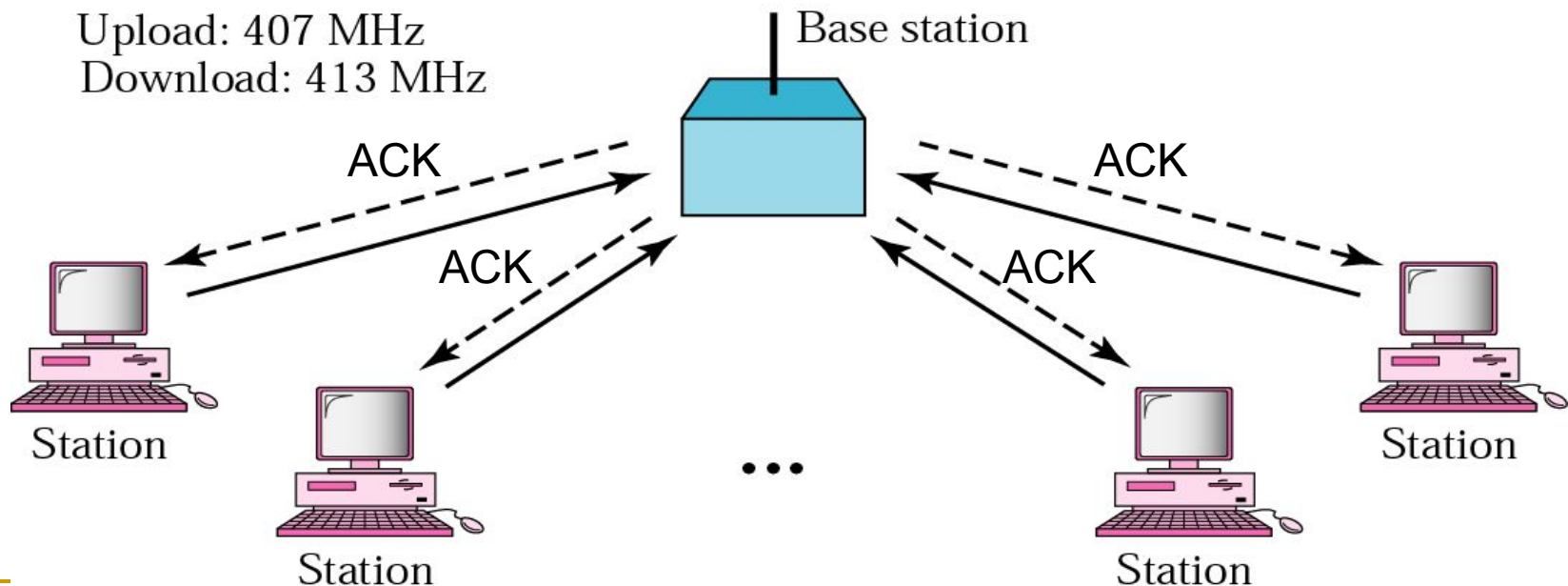


Random Access Protocols

- In **random access** or **contention** methods, no station is superior to another station and none is assigned the control over another. No station permits, or does not permit, another station to send. At each instance, a station that has data to send uses a procedure defined by the protocol to make a decision on whether or not to send.

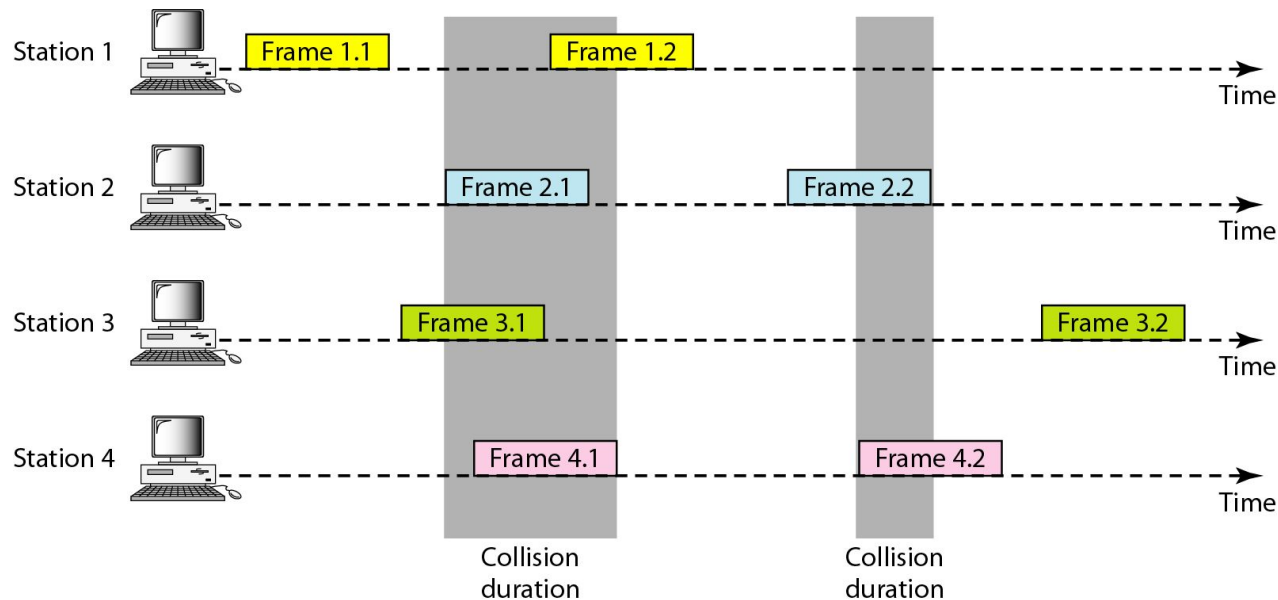
ALOHA Network

- Developed by Norm Abramson at the Univ. of Hawaii
 - the guy had interest in surfing and packet switching
 - mountainous islands → land-based network difficult to install
 - fully decentralized protocol



Pure Aloha

- The node immediately transmits its frame completely
- If the frame is collided it retransmits the frame again (after completely transmitting its collided frame) with the probability p



Slotted Aloha

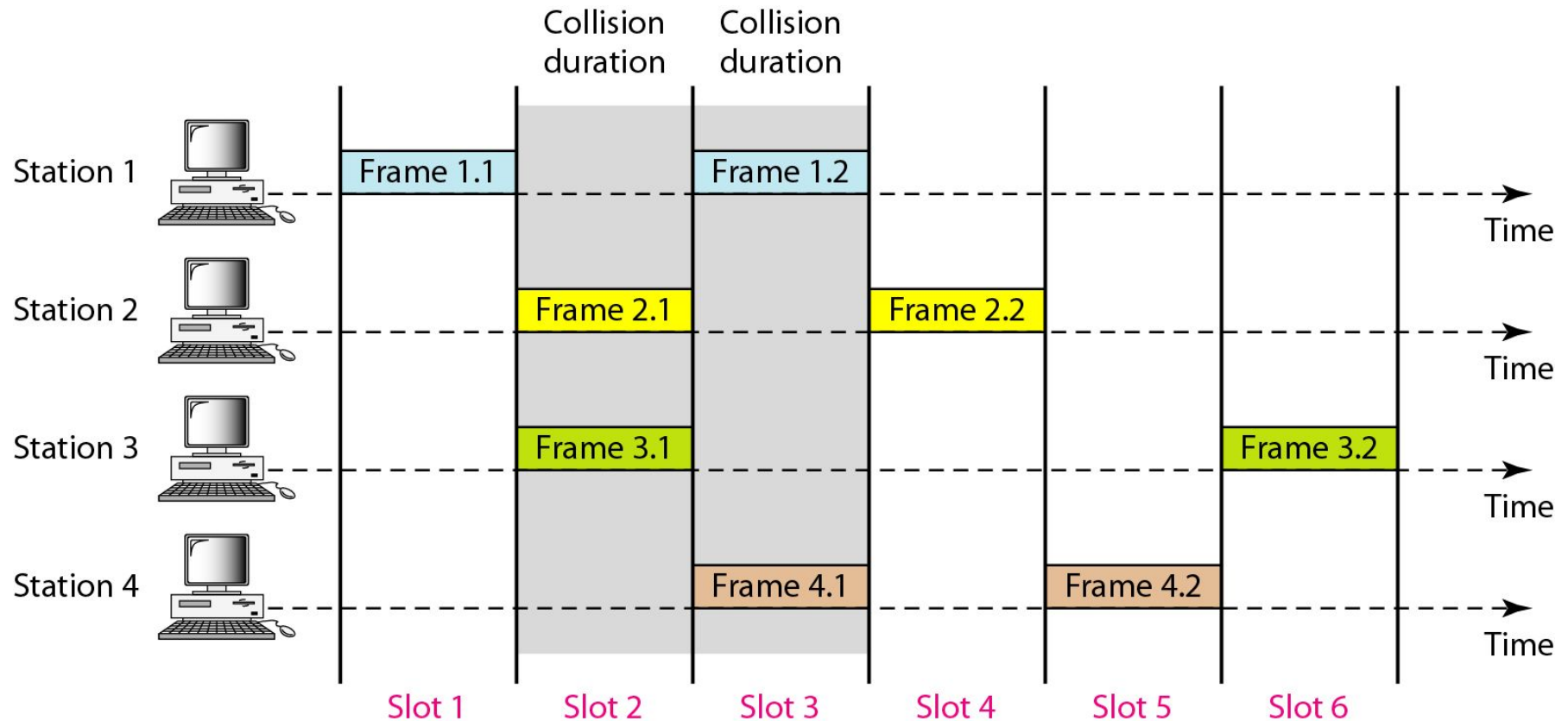
■ Assumptions

- all frames same size
- time is divided into equal size slots, time to transmit 1 frame
- nodes start to transmit frames only at beginning of slots
- nodes are synchronized
- if 2 or more nodes transmit in slot, all nodes detect collision

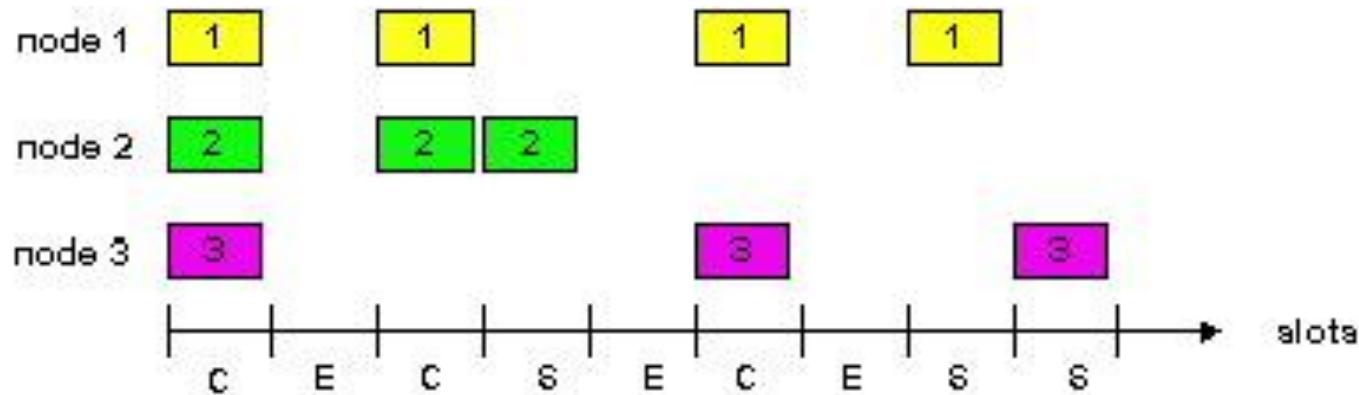
■ Operation

- when node obtains fresh frame, it transmits in next slot
- no collision, node can send new frame in next slot
- if collision, node retransmits frame in each subsequent slot with prob. p until success

Frames in a Slotted ALOHA Network



Slotted Aloha



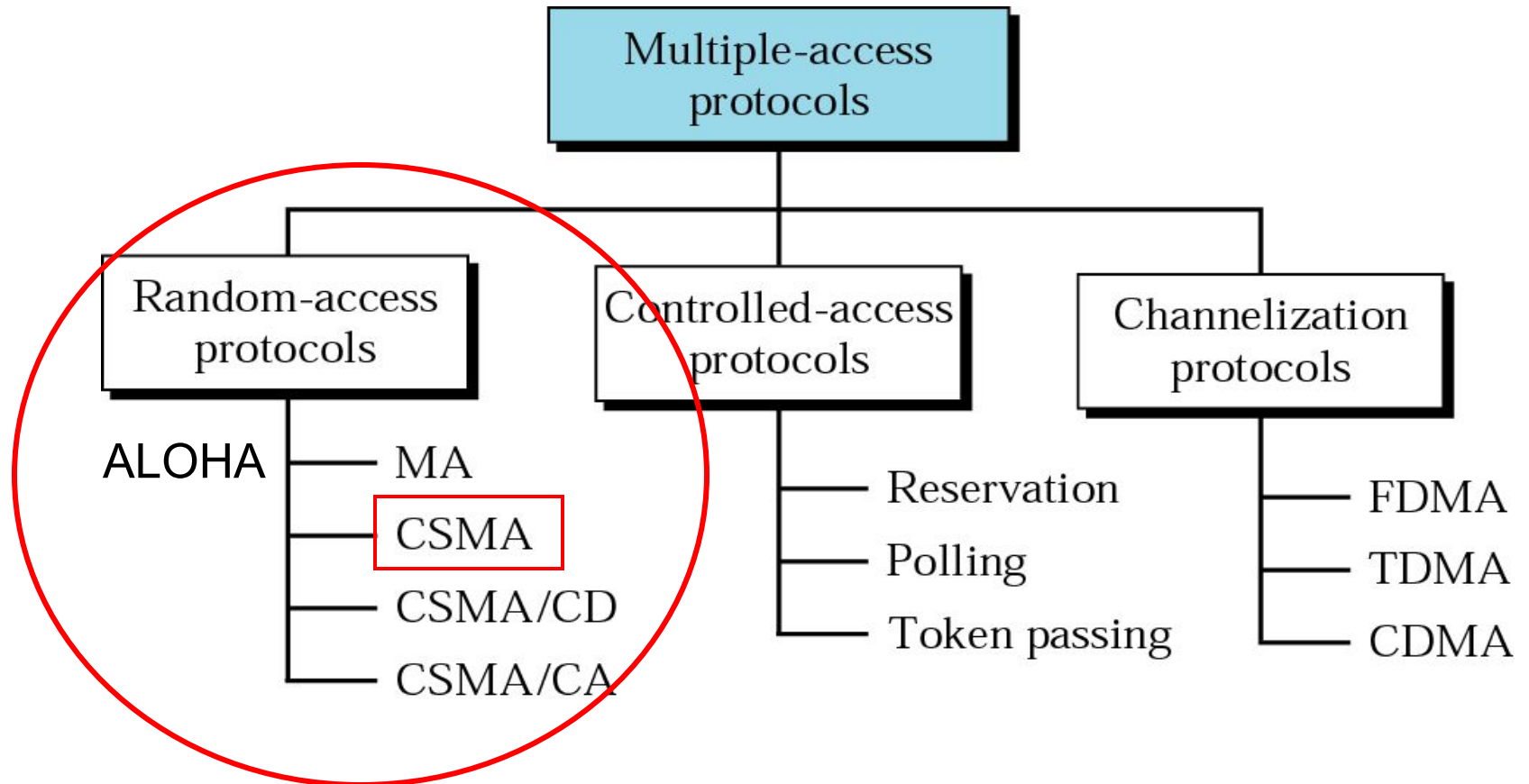
■ Pros

- single active node can continuously transmit at full rate of channel
- highly decentralized: only slots in nodes need to be in sync
- simple

■ Cons

- collisions, wasting slots
- idle slots
- nodes may be able to detect collision in less than time to transmit packet
- clock synchronization

Multiple Access Protocols

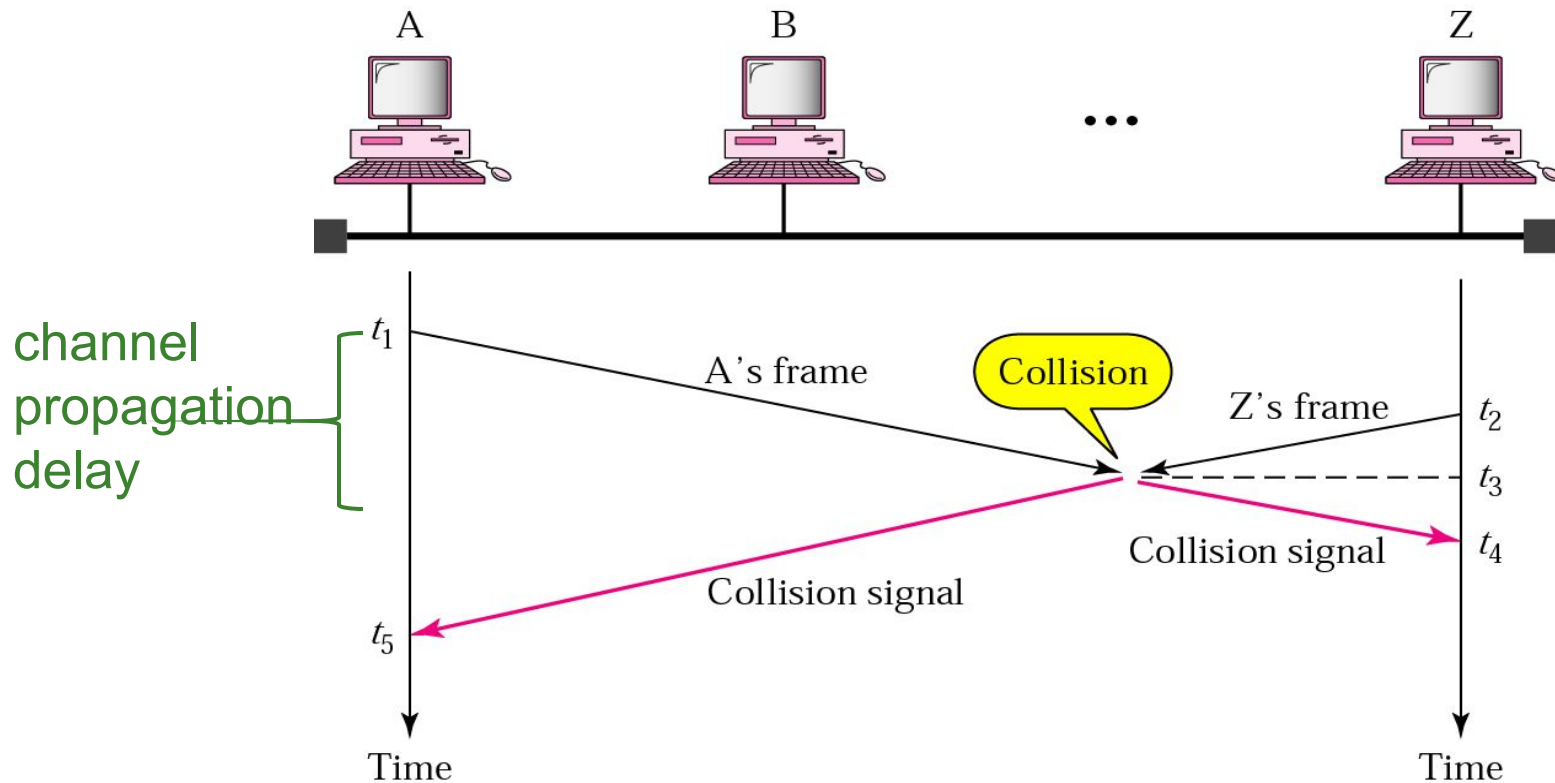


Carrier Sense Multiple Access

- Invented to minimize collisions and increase the performance
- A station now “follows” the activity of other stations
- Simple rules for a polite human conversation
 - Listen before talking
 - If someone else begins talking at the same time as you, stop talking
- CSMA:
 - A node should not send if another node is already sending
 - carrier sensing
- CD (collision detection):
 - A node should stop transmission if there is interference
 - collision detection

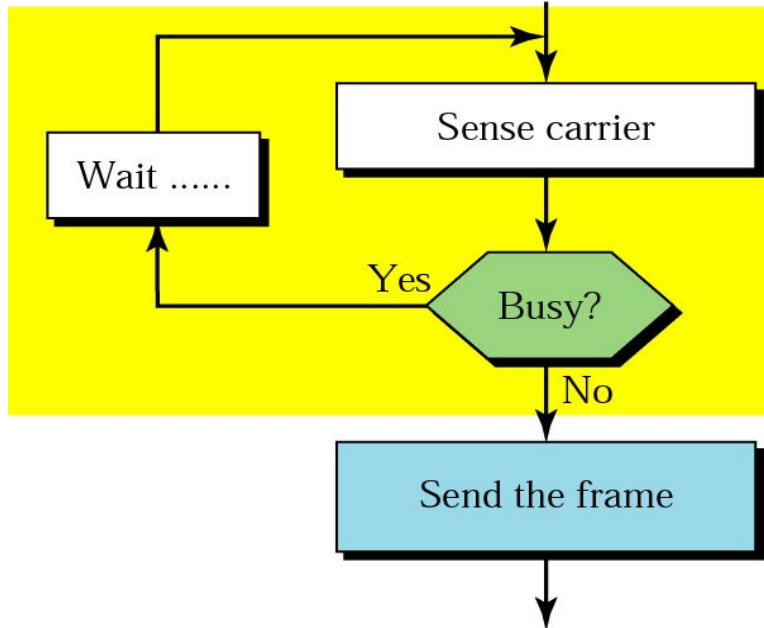
Carrier Sense Multiple Access

- If everyone is sensing the medium how come that collisions still occur?



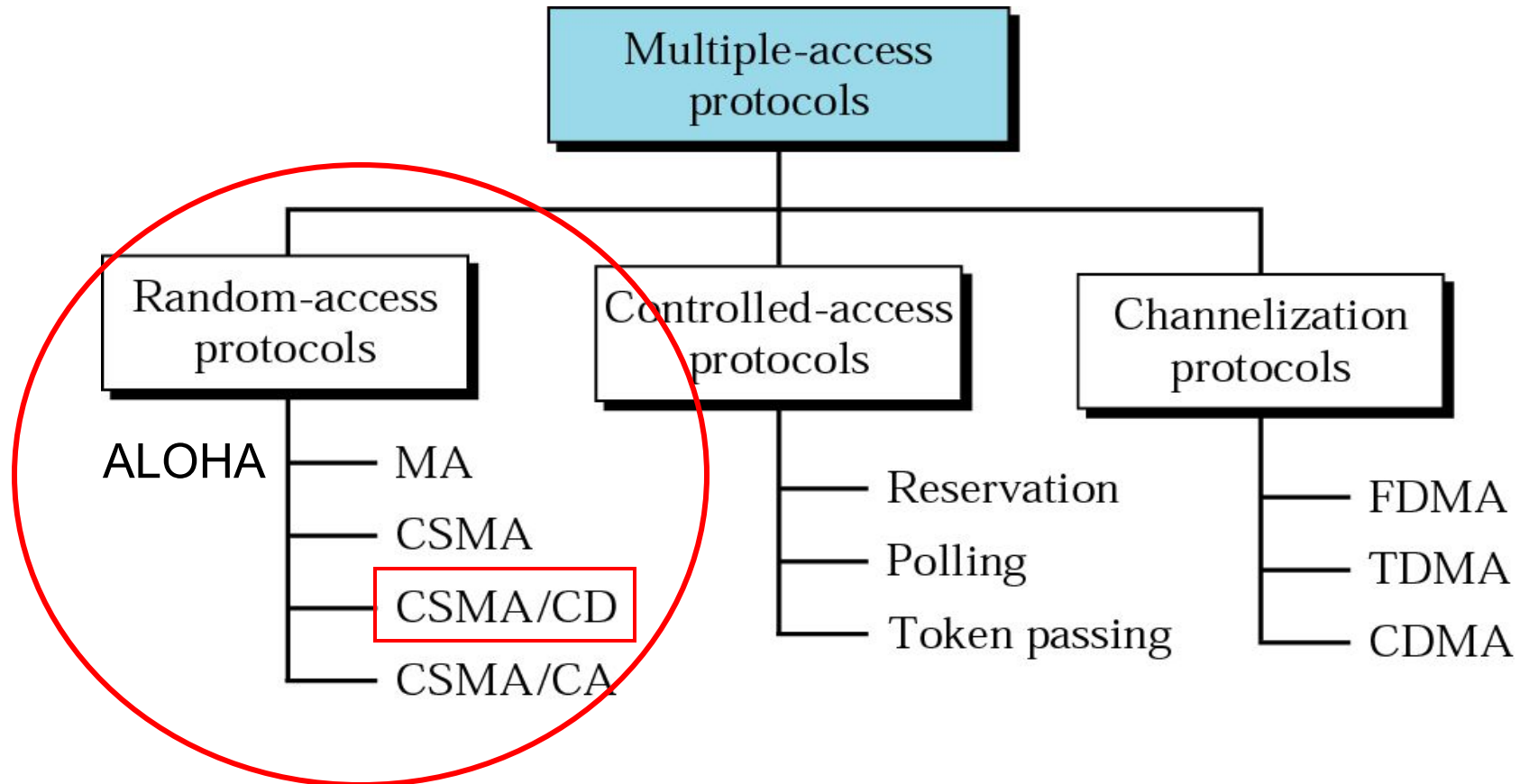
CSMA (cnt'd)

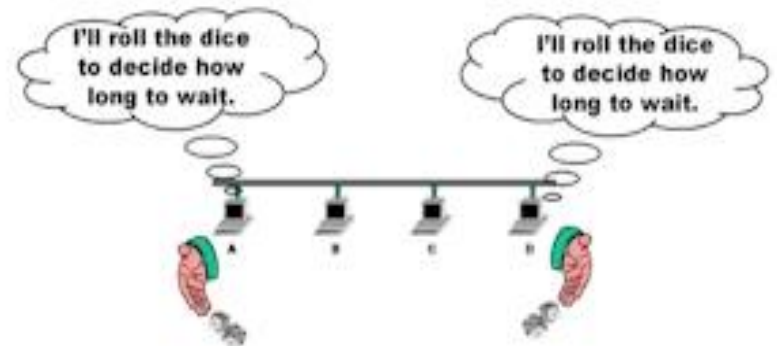
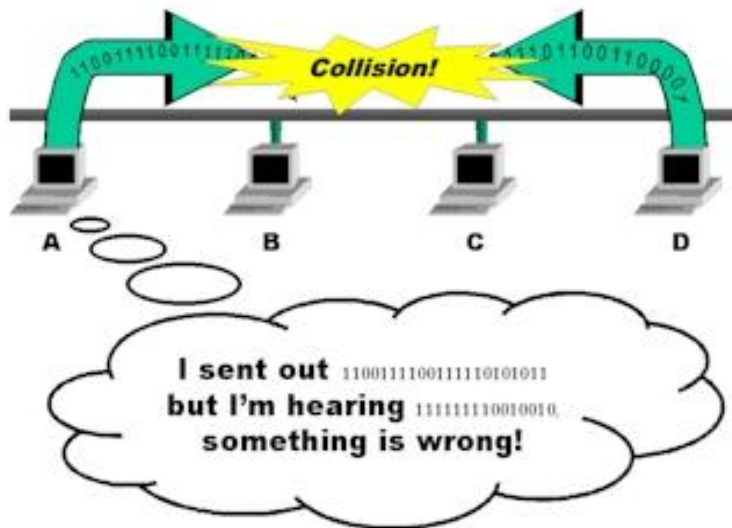
Nonpersistent strategy



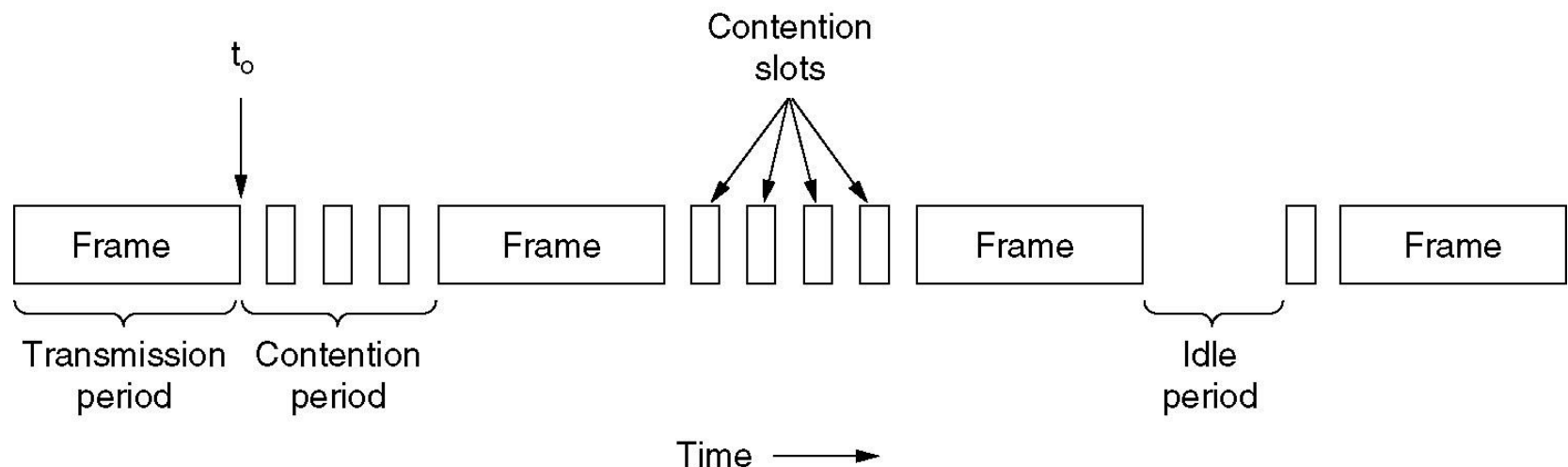
- Reduces the chance of collisions
- reduces the efficiency

Multiple Access Protocols





CSMA with Collision Detection (CSMA/CD)

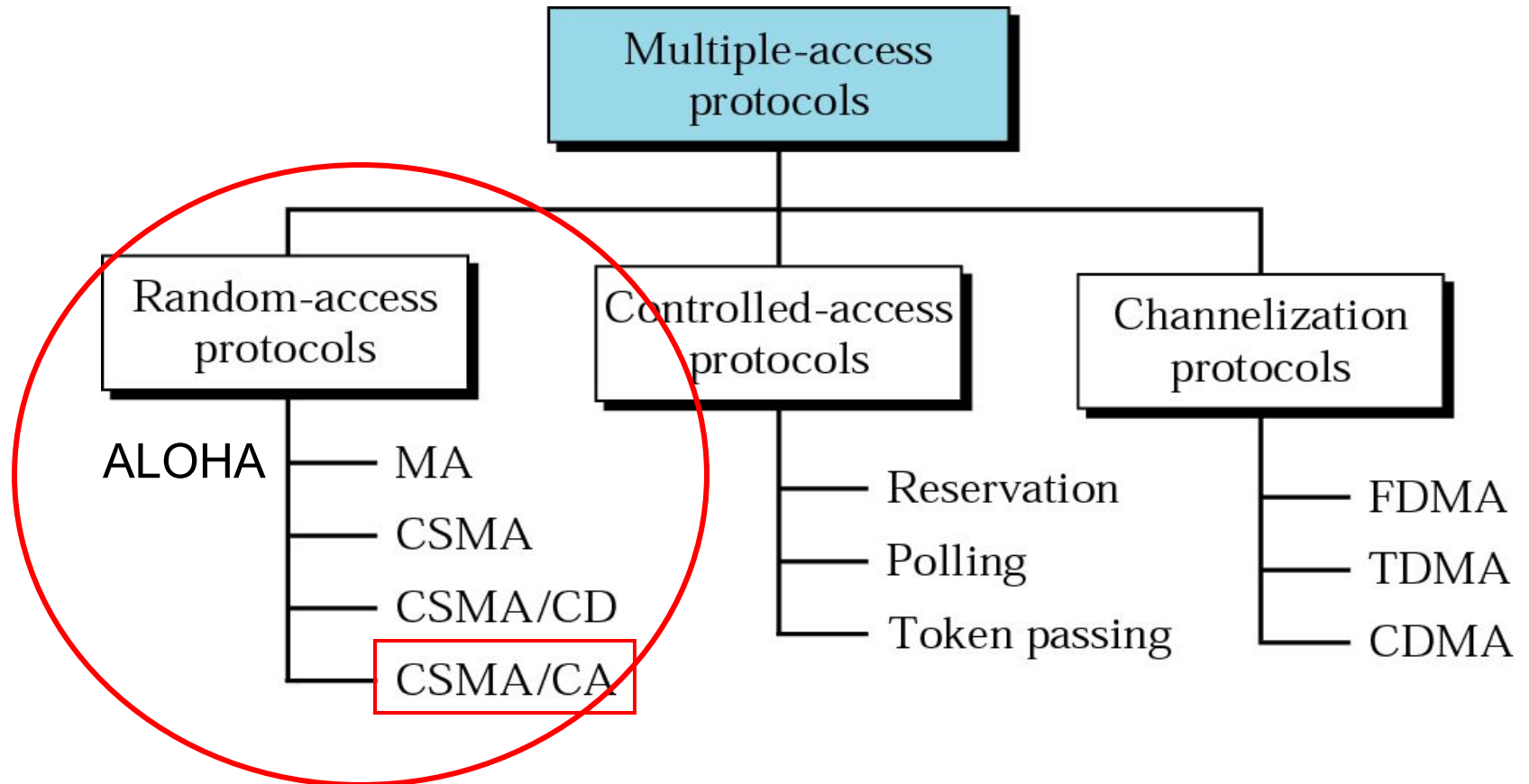


- CSMA/CD can be in one of the three states: contention, transmission, or idle.
- Example of CSMA/CD: Ethernet
- How long does it take before stations realize that there has been a collision?

Collision Detection

- How the station detects a collision?
- There are many collision detection methods!
 - Most of them are analog processes.
- Examples:
 - detecting voltage level on the line
 - detecting power level
 - detecting simultaneous transmission & reception

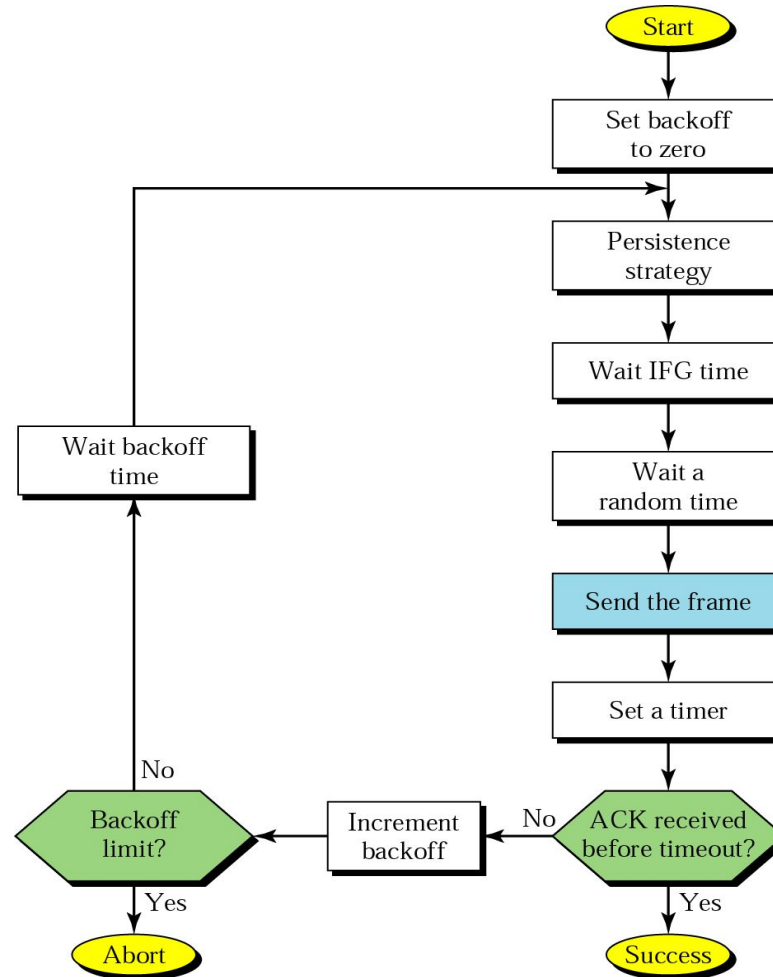
Multiple Access Protocols



CSMA with Collision Avoidance

-CSMA/CA-

- no collisions



Readings

- Chapter 12 (B.A Forouzan)
 - Section 12.1
 - (Cover only those contents which are related to topics covered in class)

